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Edinburgh Open Research Conference | Lightning Talk

Can we measure research in a more meaningful way?

A quick look at the 2025 PLOS Open Science Indicators dataset

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PLOS Open Science Indicators dataset and analysis methodology

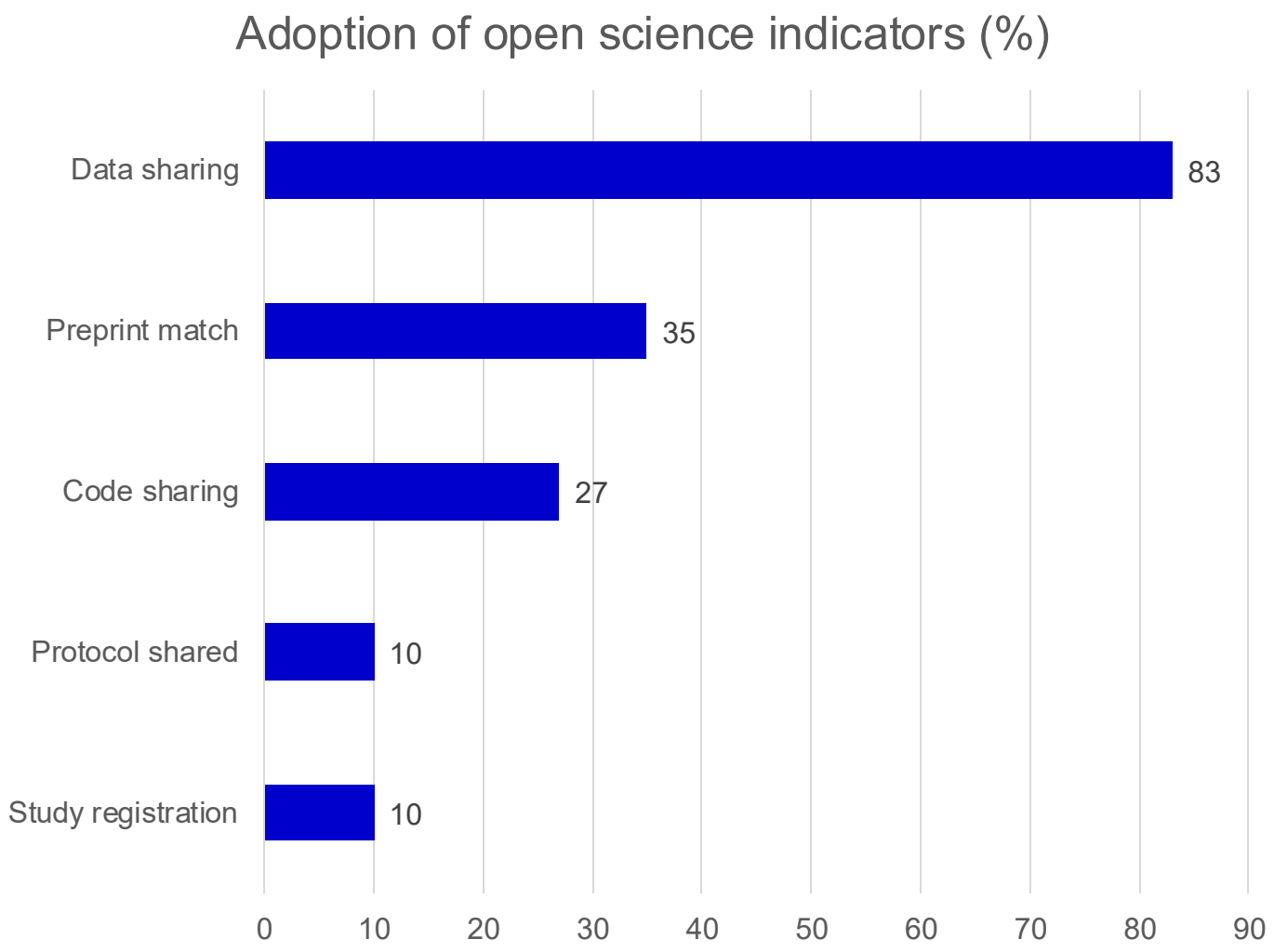
- Demonstrating different open research practices within PLOS publications
- Data availability/share, code availability/share, preprint matching, protocol sharing, and study registration
- Released since 2022 | latest version in July 2025 (including 140K outputs)

138,995
PLOS articles in the OSI dataset

1,368
Imperial articles matched by DOI

1%
Of the PLOS set and 1.45% of Imperial outputs

PLOS Open Science Indicators dataset snapshot



- **Data sharing is mature**
  - Thanks to data management policies
- **Code is the weak link**
  - Only 56% of code-generating papers share the code
- **Preprints are climbing fast**
  - From 13% in 2018 to nearly 50% in 2024
- **Protocols and registration lag**
  - 10% each. Newer practices, and more common in some fields than others

Open research practices reinforcing one another

PLOS Dataset (Imperial)	Dynamic	Default
	473 # Items	1,368 # Items

Dynamic	89.85% % Data shared (Imperial)	46.09% % Code shared (Imperial)	100.00% % Preprint match (Imperial)	11.84% % Protocol shared (Imperial)	7.40% % Registration shared (Imperial)
Default	82.75% % Data shared (Imperial)	27.05% % Code shared (Imperial)	34.58% % Preprint match (Imperial)	9.80% % Protocol shared (Imperial)	9.87% % Registration shared (Imperial)

What this means

Data sharing is strong
- High adoption of data sharing
- Data policies important

Code sharing needs a push
- Code that underpins research needs to be shared more widely

Indicators reinforce each other
- Open science indicators are not independent
- Importance of training of all principles together

Caveats

One publisher, certain disciplines

Some indicators don’t work for some disciplines

Journal-centric: works backwards from publication

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